

Networks (NW)

11 Addressing in Data Link Layer, Media Access Methods, Devices in the Network Layer

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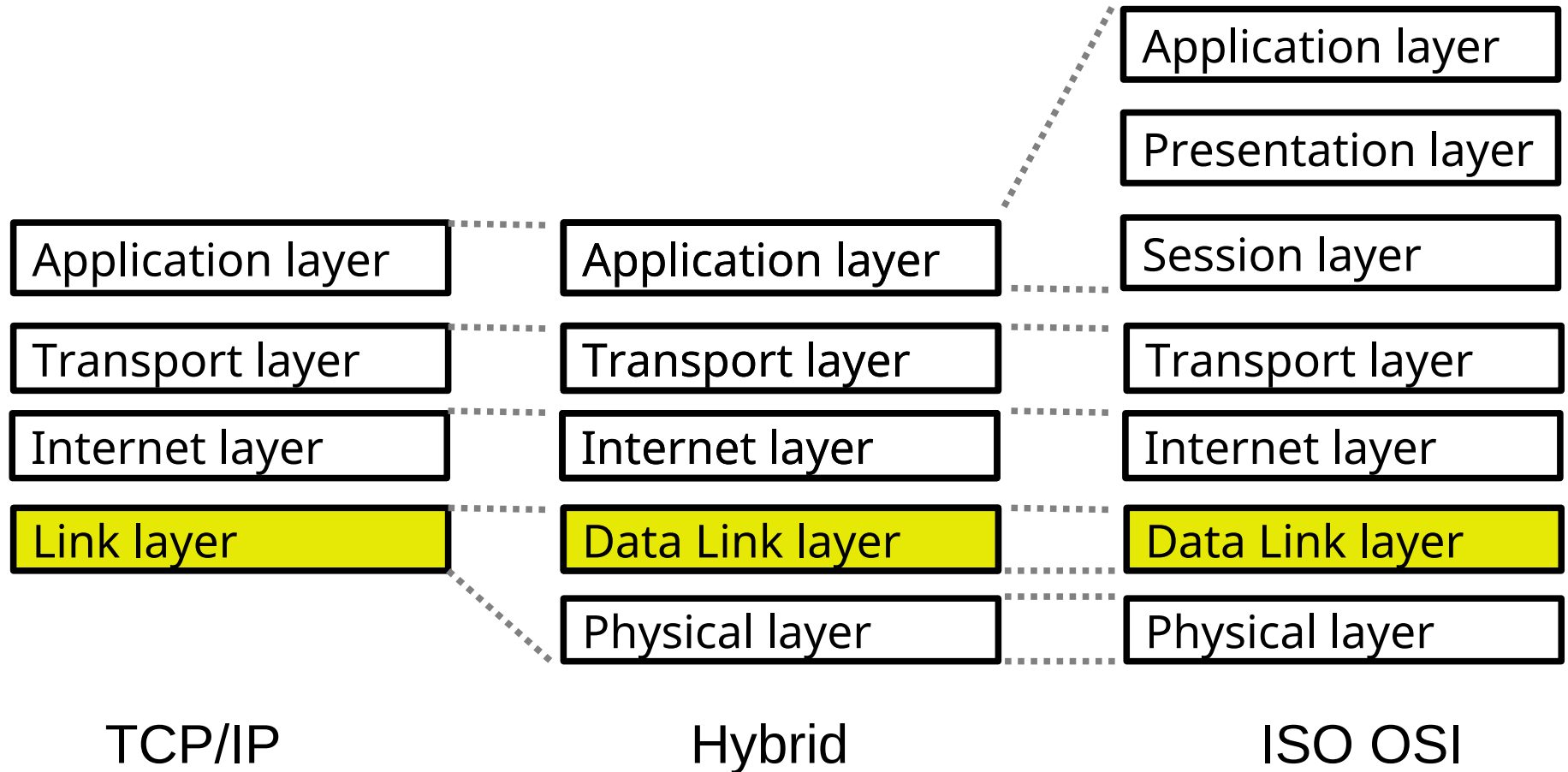
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Data Link Layer



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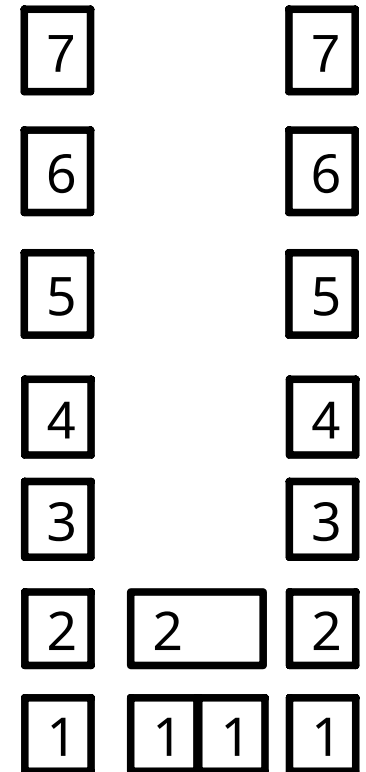


Data Link Layer

- **Tasks**
 - Packet shaping from data of network layer (sender)
 - Correct data bundling from frames of physical layer (receiver)
 - Correct transmission of data frames
 - Error detection, error correction by checksum
 - Provision of **physical addresses**
 - Access control according to **media access schemes**
- **Devices working (only) in this layer**
 - Bridge, Layer-2-Switch (Multiport Bridge), Modem
- **Typical protocols: Ethernet, WLAN, Bluetooth, PPP**

Devices: Bridge, Layer-2-Switch

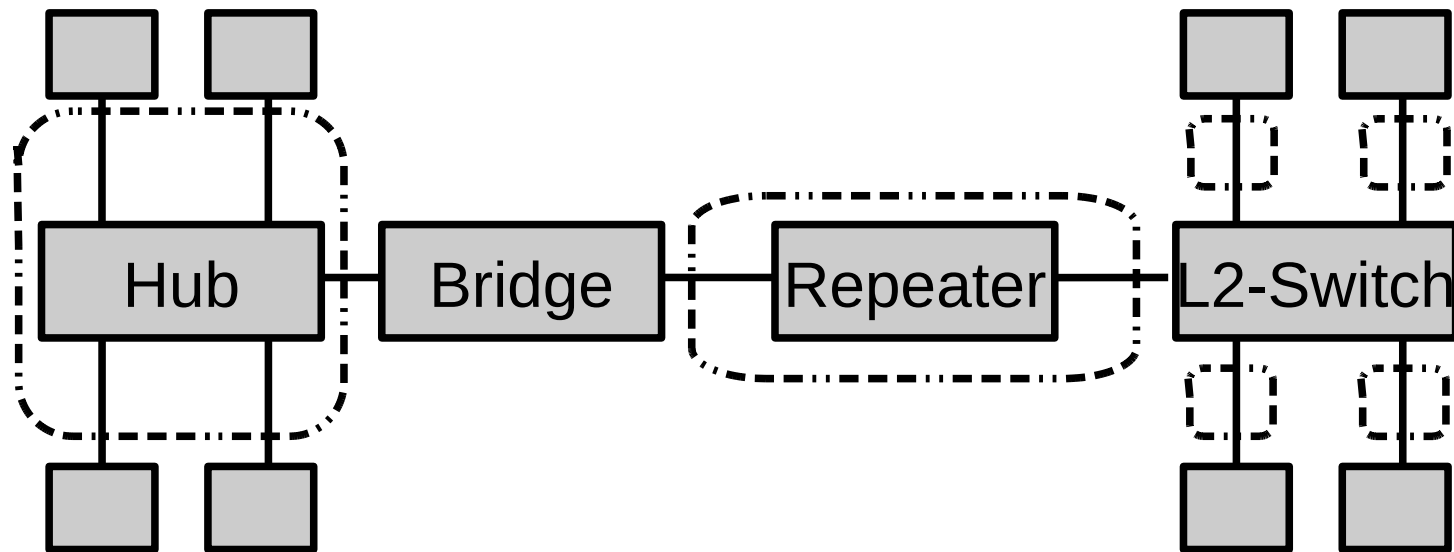
- **Bridge** connects 2 networks (typically with different transmission media)
- **Layer-2-Switch** aka **multiport bridge** connects > 2 networks
- **Goal:** increasing length of network distance span
- No filtering and forwarding of frames
→ no active participation in network communication
- By bridge/layer-2-switch separated networks
separate collision domains



Collision Domains



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Collision Domain

Media Access Control (MAC) Addresses

6 Bytes (= 48 bits) → 2^{48} unique addresses possible

- **Given in hexadecimal, each byte separated by dashes or colons**
 - Example: **90:8d:6e:2f:ea:e0**
- **Broadcast address: ff:ff:ff:ff:ff:ff**
 - Data frame to this address reaches all devices in the network
- **Permanent to each network device (sometimes SW adjustable)**
- **How to show unter Linux:**
 - `>ip link show`

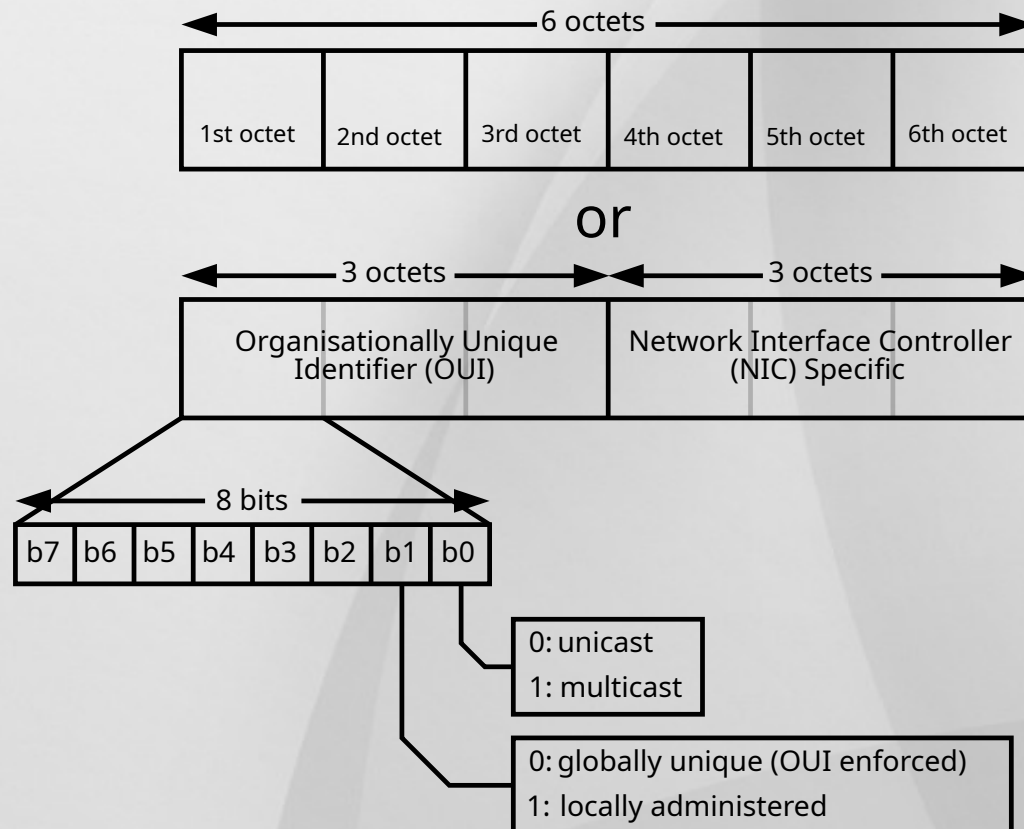
How to Ensure Uniqueness of MAC-Addresses?



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- **Managed** by Institute of Electrical and Electronics Engineers (**IEEE**)
- **MAC Address Block Small (MA-S), Medium (MA-M), Large (MA-L)**
- **MA-L allow companies to use 24 Bit as individual addresses per Network Interface Controllers (NICs)**
- **MA-L previously referred to as Organizationally Unique Identifier (OUI)**
- **e.g. from <https://www.wireshark.org/tools/oui-lookup.html>:**
 - 90:8D:6E Dell Inc.
 - 00:04:0F Asus Network Technologies, Inc.
 - D4:F3:2D Intel Corporate

Multicast and Private MAC Addresses

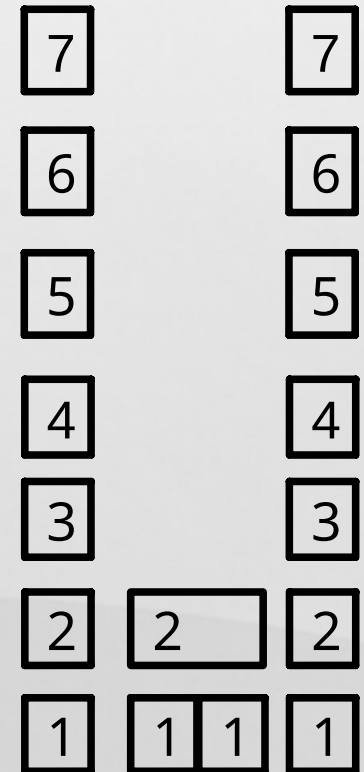


- **Locally administered MAC addresses are used for virtualization (containers and virtual machines)**
 - e.g.: ae:1c:c8:ce:32:6a
- **Multicast Addresses are needed for Spanning Tree Protocol (STP) and IPv6**

Source: https://en.wikipedia.org/wiki/MAC_address#/media/File:MAC-48_Address.svg

Addressing on Data Link Layer

- **Layer 3: Logical Addresses (e.g. IP Addresses)**
- **Layer 2: Physical Addresses**
- **Requires a „Mapping“ Protocol to match logical and physical addresses, e.g. in Ethernet Address Resolution Protocol (ARP) or Neighbor Discovery Protocol (NDP)**

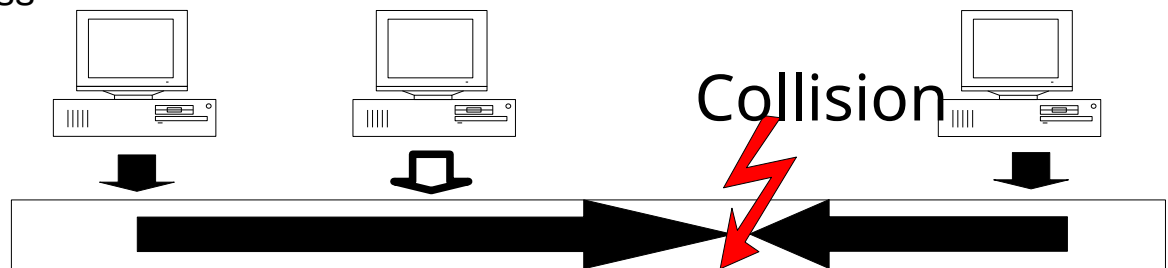


Security Aspects of MAC Addresses

- **MAC address filters are often used (check your DSL-Router config at home)**
- **MAC addresses are visible to all network participants**
- **Security level is low, because MAC addresses can be mostly configured by Software**
 - > ip link set dev <Interface> address <MAC Address>**
- **Thread of MAC address spoofing (especially in WiFi-Networks)**

Reasons for Media Access Control

- Multiple terminals accessing a network requires coordination of shared resource network in a collision domain (see before)
- **Problem:**
 - Sending of data by two or more terminals on the carrier at the same time possible aka „Collisions of data“
 - No master to coordinate access



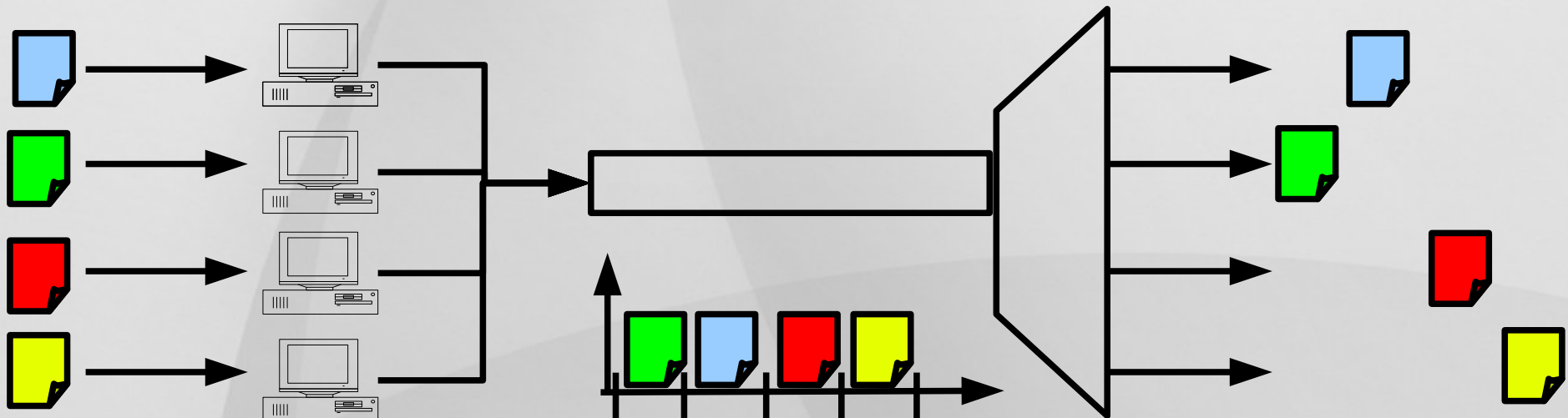
- **Solution:** Media access control schemes
 - Collision Avoidance (CA)
or
 - Collision Detection (CD)
- **Goal:**
 - Minimizing data re-transmissions to minimize waiting time (increased latency)

Media Access Control

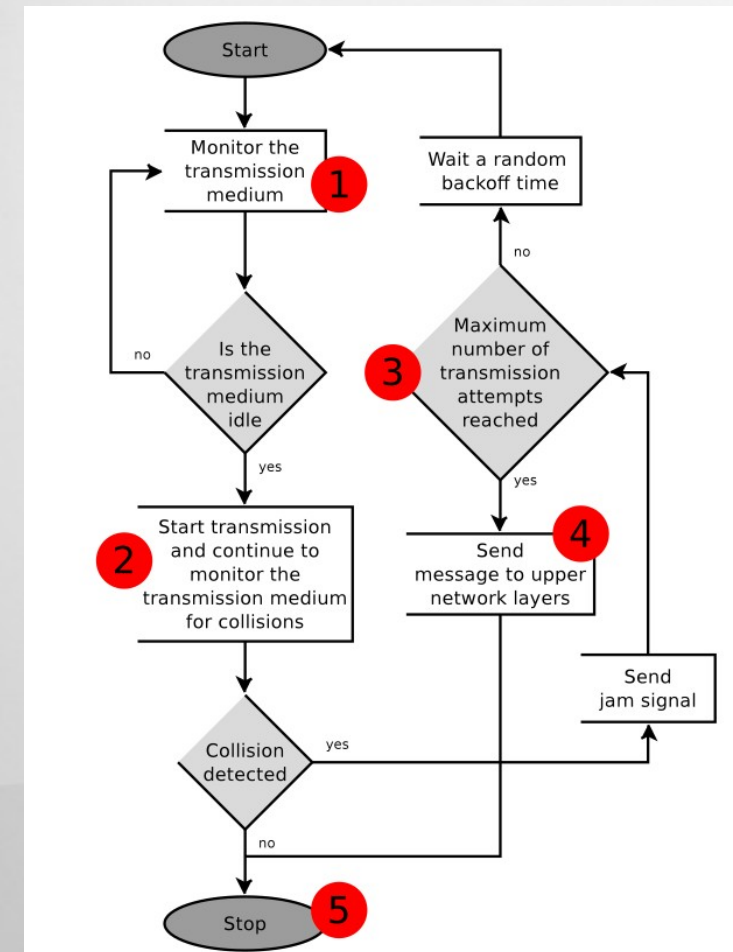
- **Carrier Sense Multiple Access (CSMA)**
 - Collision Detection (CD) → Ethernet
 - Collision Avoidance (CA) → WLAN
- **Time Division Multiple Access (TDMA)**
 similar to Round Robin in Process Scheduling

Time Division Multiple Access (TDMA)

- „Each frame at a given time“
- **Required:**
 - Each terminal „knows“ the exact time to send its messages
 - Exact time distribution
- **GSM/UMTS and other Mobile Communication Standards**



- **Back off time:**
Random waiting time
 - Minimum value $< t <$ maximum value of Contention Window alue (CW)
 - Exponential growth
- **Maximum numbers of transmission attempts ≤ 16**
- **Absolutely necessary in Ethernet networks with bus topology**
 - Becomes rare due to trend towards fully switches networks (star topology with Layer-3-Switches)



Source: C. Baun, Computer Networks, Springer Vieweg 2025

CSMA/CA, Collision Avoidance (CA)

- **Node sends a frame after a Distributed Interframe Space (DIFS) time period and receiver acknowledges its reception after a Short Interframe Spacing (SIFS) time period**
- **Other Terminals have to wait sending for a DIFS and random backoff time**
- **Problems with Hidden Terminals → Solution RTS/CTS**

Applications:

- **WIFI (802.11), Apple Talk**
- **Controller Area Network (CAN)**

CSMA/CA Variants

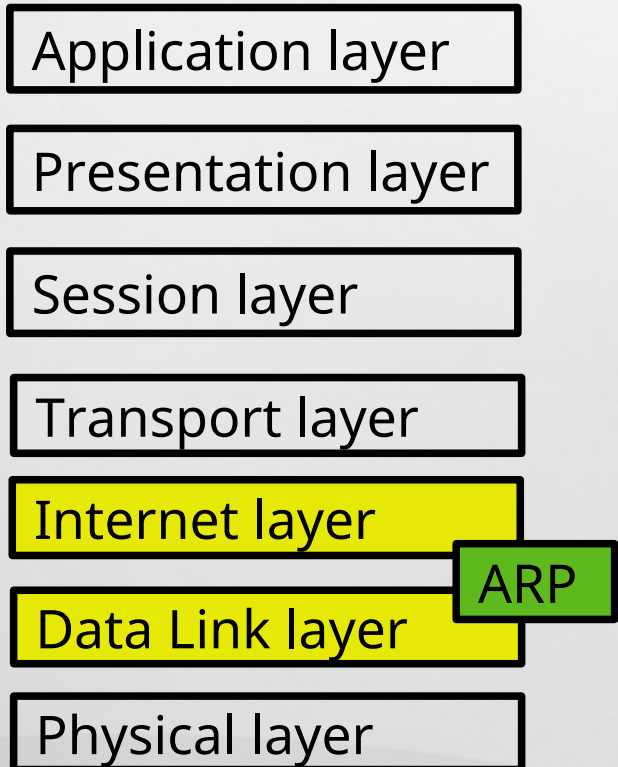
- **Request To Sent/Clear To Sent (RTS/CTS)**
 - Node with sent request signals sending intention
 - Pre message, to signal intention for sending
 - Pre message has to be received by all nodes (ACK needed)

Drawbacks:

- Additional communication overhead by sending intention
 - Communication rate drops & Waiting time increases latency
- **Point Coordination Function (PCF)**
 - Access Point coordinates access to the transmission media
 - All terminals must register to AP
 - AP asks terminals to transmit
 - Rarely implemented

Address Resolution Protocol (ARP)

- **Network Layer uses Internet Protocol (IP) Addresses**
- **Data Link uses MAC Addresses**
- **Address Resolution needed on Data Link Layer**
 - ARP sends a MAC broadcast FF:FF:FF:FF:FF:FF with requested IP address
 - Network device with the IP address replies with MAC address
 - IP address and MAC address are stored in **ARP cache** of the requestor
 -

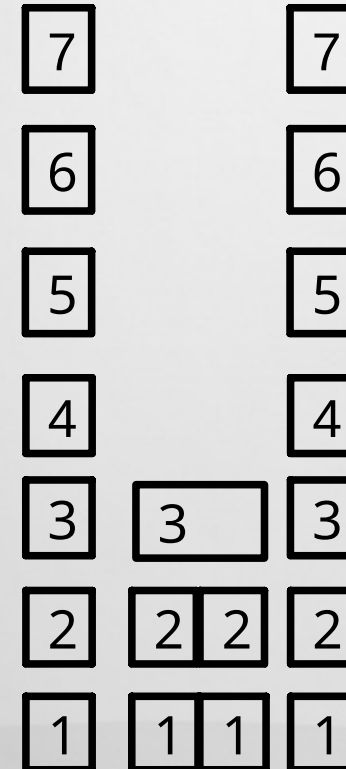


ARP Cache

- **Goal: Speeding up the address resolution**
- **Cache table contains:**
 - Protocol type (IP)
 - IP address
 - MAC / HW address
 - Time To Live (TTL) – timespan for entry living (typ. 5Mins)
- **Commands to print ARP cache contents**
 - >arp -n
 - >ip neighbor

Devices on the Network Layer

- **Router & Layer-3-Switches**
 - Forward packets from networks with **different logical address ranges**
- **Router (in Addition):**
 - Could connect a LAN to a WAN (e.g. DSL- or 4G/5G-Router)
- **Gateways are protocol converters**
 - On layer 3 (or 4)
 - e.g. Ethernet to ISDN, VPN gateway
- „Default gateway“ refer to a device
 - which (formerly) converts protocol
 - still used even if this device is a router



Broadcast Domain

- **Logical part of a computer network**
- **Broadcasts reach all devices / terminals in that domain**
- **Router/Layer-3-Switch separate broadcast domains**
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